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Multimodal-guided self-distillation for unified person search

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ABSTRACT

Person search is challenging due to limitations in identity representation. Existing methods rely on one-hot encoding, ignoring semantic relationships among pedestrians. This leads to a fragmented feature space and reduces generalization ability, especially in large-scale scenarios with a significant proportion of unlabeled identities. For instance, in the CUHK-SYSU dataset, 72.7% of pedestrians lack identity annotations, limiting the effectiveness of supervised learning. To address these issues, we propose a novel Multimodal-Guided Self-Distillation (MGSD) method for Unified Person Search that leverages multimodal textual descriptions and self-distillation to enhance pedestrian representation learning. Specifically, we introduce three key innovations: (1) *Multimodal LLM-Assisted Text Generation (MLTG)* to provide fine-grained semantic context beyond discrete identity labels, enabling the model to capture inter-person relationships based on clothing attributes, appearance features, and environmental cues; (2) *Semantic Structural Consistency Constraint (SSCC)* to impose global structural constraints on the feature space, ensuring that distinct identities remain separable while preserving semantic similarities among visually similar individuals; and (3) *Multimodal-Aware Self-Distillation Framework (MSDF)*, where the learnable visual encoder is progressively aligned with the pre-trained CLIP multimodal encoder, improving robustness to variations in illumination, occlusion, and background clutter. Extensive experiments demonstrate that our method significantly enhances retrieval accuracy and generalization, achieving state-of-the-art performance with an mAP of 56.1% on the PRW dataset while maintaining computational efficiency for large-scale real-world applications.

1. Introduction

Person search, which combines both person detection and person re-identification (ReID), has critical applications in areas such as intelligent surveillance, autonomous driving, and public safety. Compared with traditional ReID, person search not only requires accurate detection of target pedestrians in complex scenes but also demands retrieval of all instances of the same identity from large-scale galleries, resulting in greater challenges. Although recent methods have demonstrated impressive performance, existing models still suffer from limited generalization and inadequate feature representation in real-world scenarios, making them unstable in the presence of diverse contexts and large amounts of unlabeled data.

One principal cause of these limitations lies in the simplistic representation of pedestrian identity. Current mainstream approaches often employ one-hot encoding to represent identities, viewing each person as an independent discrete category. As shown in Fig. 1a, while this encoding highlights the uniqueness of an identity, it fails to capture the semantic relationships among different individuals, such as appearance, clothing, and contextual information, leading to an overly discrete fea-

ture space that lacks structured relationships. Moreover, popular person search datasets often suffer from limited labeled data due to incomplete identity annotations. For example, in the CUHK-SYSU dataset, approximately 72.7% of the pedestrians lack identity labels, and in the PRW dataset, about 17.4% are unlabeled. For these unlabeled pedestrians, previous person search methods mostly rely on one-hot encoding-based supervision like the Online Instance Matching (OIM) loss (Xiao et al., 2017) and struggle to provide effective guidance, thereby limiting their scalability in real-world settings with abundant unlabeled data.

Recent progress in vision-language models such as CLIP has opened up new possibilities for enhancing person-centric tasks. Several CLIP or LLM-based approaches (e.g., CLIP-ReID Li et al., 2023, PLIP Zuo et al., 2024, TBPS-CLIP Cao et al., 2024) have demonstrated the potential of multimodal pre-training to improve pedestrian representation learning by leveraging rich semantic context. However, existing text alignment-based methods (Cao et al., 2024; Jiang & Ye, 2023; Ke et al., 2024; Li et al., 2023; Liu et al., 2025; Zhang et al., 2025; Zuo et al., 2024) primarily focus on instance-level matching (e.g., minimizing distribution discrepancies between individual visual features and CLIP-generated text embeddings), which frequently leads to feature misalignment in

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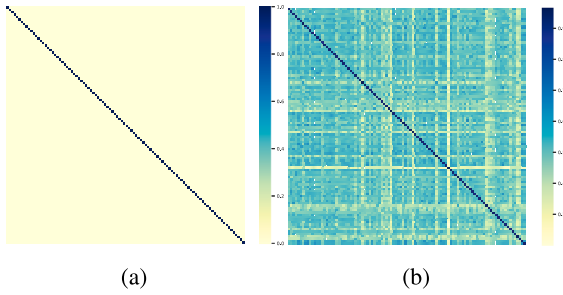


Fig. 1. Person relationship under different encoding methods. (a) Person relationship based on one-hot encoding. (b) Person relationship based on textual descriptions.

complex scenarios. In situations with numerous visually similar individuals, merely aligning each pedestrian independently to textual descriptions risks fragmenting the feature space and neglecting the broader semantic relationships among pedestrians, thereby weakening discriminative power and robustness. Furthermore, bridging these modalities remains a fundamental challenge due to modality heterogeneity, where textual semantics and visual pixels reside in disparate feature spaces, making robust alignment and cross-modal consistency inherently difficult to achieve.

To address the above issues, we propose a novel method, **Multimodal-Guided Self-Distillation for Unified Person Search (MGSD)**, which leverages multimodal textual descriptions and self-distillation to enhance pedestrian representation learning. As illustrated in Fig. 1b, we introduce **Multimodal LLM-Assisted Text Generation (MLTG)** to derive semantic context beyond discrete identity labels. Unlike traditional person search methods that rely on one-hot encoding, these textual descriptions provide rich and fine-grained semantics, such as clothing attributes, appearance features, and contextual environmental cues, enabling the model to explicitly capture semantic inter-person relationships.

To further enforce semantic coherence in the learned embedding space, we introduce a **Semantic Structural Consistency Constraint (SSCC)** module, which leverages text-based supervision to impose global distribution constraints on the feature space. While prior multimodal alignment methods focus solely on matching individual image-text pairs, our constraint ensures distinct identities remain clearly separable while simultaneously preserving the latent semantic similarities among visually similar individuals, thus maintaining the structural integrity of the high-dimensional representation space.

Moreover, to enhance robustness under real-world variations, such as changes in illumination, occlusion, and background clutter, we introduce a **Multimodal-Aware Self-Distillation Framework (MSDF)**. Different from previous distillation strategies that focus on simple feature mimicking and often lack the integration of rich multimodal semantic information, we employ the pre-trained multimodal CLIP encoder as a stable knowledge source, progressively aligning the learnable visual encoder with this multimodal prior during training. This self-distillation process significantly bolsters the model's generalization ability.

By integrating these innovations, our proposed MGSD framework effectively captures inter-person semantic relationships, substantially enhancing retrieval accuracy and generalization across diverse environments. Extensive experiments validate our approach, achieving state-of-the-art performance with an mAP of 56.1% on the PRW dataset while ensuring computational efficiency suitable for large-scale, real-world applications.

In summary, our approach tackles the fundamental limitations of existing person search methods by:

- Utilizing multimodal textual descriptions to overcome the limitations of one-hot encoding and establish a more informative pedestrian relationship representation.

- To better capture inter-person relationships, we introduce a semantic correlation consistency loss, which enforces structural constraints on the feature space and improving retrieval robustness and discriminability in large-scale person search tasks.
- We integrate CLIP's pre-trained multimodal knowledge into a self-distillation framework, where a learnable visual encoder is progressively aligned with the frozen CLIP encoder. This enhances the model's robustness to illumination variations, occlusions, and background changes.

2. Related work

2.1. Person search

Person search is an extension of the person ReID (Dai et al., 2025; Jia et al., 2025b; Wang et al., 2024a,c,d; Wu et al., 2020; Yin et al., 2023; Zhu et al., 2024) task, which aims to search for and locate persons in scenes. Most existing methods can be divided into two-step (Chen et al., 2020b; Dong et al., 2020b; Wang et al., 2020; Yao & Xu, 2020) and end-to-end approaches. The two-step methods use two independent networks to generate accurate person features and positioning information, effectively avoiding the conflict between detection and ReID features. DPM (Zheng et al., 2017) is a pioneering effort that introduces a large-scale person search benchmark dataset. It systematically evaluates various combinations of detectors and recognition modules, ultimately proposing a confidence-weighted similarity method to suppress false positive targets effectively. Building upon this foundation, TCTS (Wang et al., 2020) introduces an identity-guided query detector, elevating query quality and mitigating non-query target proposals. To substantially reduce repeated proposals, IGPN (Dong et al., 2020b) contributes to the Siamese region proposal network, facilitating learning of the similarity between queries and proposals.

Compared to two-step approaches, end-to-end (Cao et al., 2022; Chen et al., 2021; Han et al., 2021a; Jia et al., 2024; Yan et al., 2023; Yu et al., 2022) methods unify two tasks in the same framework for better efficiency. Yan et al. (2023) propose the anchor-free person search method for the first time and try to alleviate the conflicts caused by the two tasks from different aspects. Cao et al. (2022) introduce a person search training mechanism based on one-to-one matching, learning more discriminative features, and eliminating NMS post-processing operations, significantly improving network reasoning speed. In addition, weakly supervised person search methods (Han et al., 2021b; Jia et al., 2023) aim to train models using only bounding box annotations, without requiring identity labels. Unsupervised domain-adaptive approaches (Li et al., 2022b; Wang et al., 2024b) further explore transferring pre-trained models from labeled source domains to unlabeled target domains, mitigating the need for manual annotation in new environments. In this paper, we adopt SeqNet paradigm and enhance task consistency across different tasks.

2.2. Multimodal vision-language learning

Recent advancements in Large Language Models (LLMs) (Chowdhery et al., 2023; Yang et al., 2023a; Zheng et al., 2023) have driven the rise of Multimodal Large Language Models (MLLMs) (Liu et al., 2023; Touvron et al., 2023; Zhu et al., 2023), which integrate text and non-textual data (e.g., images) by mapping them into a shared space and leveraging a language decoder. Refinements in token alignment and decoding (Alayrac et al., 2022; Li et al., 2024a) have further stabilized multimodal training. In addition, CCR-Net (Wu et al., 2021) addresses the fusion of heterogeneous data (e.g., hyperspectral and LiDAR) by designing a cross-channel reconstruction module that mutually predicts multimodal features for highly compact representation learning. Similarly, FlexiMo (Li et al., 2026) develops a spatial resolution-aware foundation model to adaptively recalibrate patch embeddings across arbitrary spatial resolutions and diverse sensor specifications. Beyond

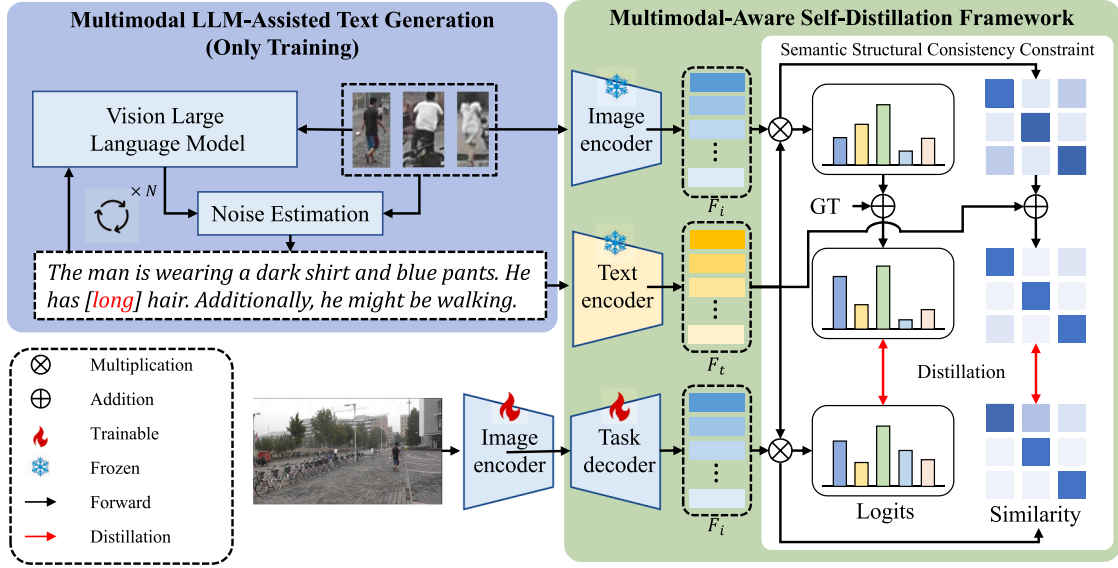


Fig. 2. Overview of our proposed framework. We first utilize a Multimodal LLM-assisted Text Generation (MLTG) module, leveraging a fixed-instruction prompt and CLIP-based noise estimation for stable and accurate pedestrian descriptions. A Semantic Structural Consistency Constraint (SSCC) is then introduced to preserve meaningful inter-person relationships and structural integrity within the feature space. Subsequently, a Multimodal-Aware Self-Distillation Framework (MSDF) integrates semantic-aware multi-distillation to ensure consistency between image and text representations, thereby enhancing the generalization and robustness of the model.

MLLMs, vision-language pre-training (VLP) has advanced various tasks by aligning images and text, as seen in CLIP (Radford et al., 2021). Further developments expand training objectives with masked modeling, image-to-text matching, and cross-modal fusion (Li et al., 2022a). Moreover, MambaX (Li et al., 2025) implements a nonlinear state-space AI framework to regulate multimodal feature evolution and reconstruction through a novel cross-control paradigm. Methods like ALBEF (Li et al., 2021) and SimVLM (Wang et al., 2021) enhance alignment and cross-modal attention, while tuning techniques (e.g., CoOp Zhou et al., 2022b, CoCoOp Zhou et al., 2022a) specialize models for tasks like classification, segmentation, and detection. In fine-grained ReID, CLIP-based approaches have recently emerged. CLIP-ReID (Li et al., 2023) optimizes learnable text tokens with frozen CLIP encoders before fine-tuning the visual encoder. PLIP (Zuo et al., 2024) pre-trains a person-centric model with tasks like text-guided colorization and attribute prediction to enhance fine-grained feature representation. Focusing on data augmentation and loss function optimizations, TBPS-CLIP (Cao et al., 2024) empirically investigates the use of CLIP for text-based person search to enhance retrieval efficiency and accuracy within a lightweight architecture. In contrast, MGSD addresses the unified person search task by jointly leveraging LLM-generated textual semantics and CLIP-based distillation within a single multimodal framework. Unlike prior methods that rely on fixed prompts, pre-training, or empirical tuning, our approach enables adaptive semantic supervision and global relational consistency learning, effectively bridging detection and ReID through structured multimodal interaction. Similarly, DCFL (You et al., 2025) enhances fine-grained cross-modal alignment by constructing co-saliency features through intra-modal saliency learning and cross-modal saliency matching, while further promoting feature diversity via a regularized mining strategy.

3. Method

The overview of our proposed MGSD framework is illustrated in Fig. 2. First, we utilize MLTG, leveraging LLMs (Yang et al., 2024) to generate rich semantic descriptions of pedestrians. These descriptions are then refined through CLIP-based noise estimation. To preserve meaningful inter-person relationships and structural integrity within the feature space, we introduce a SSCC. Finally, MSDF progressively aligns the vi-

sual encoder with a pre-trained CLIP multimodal encoder, enhancing robustness and generalization across diverse datasets.

3.1. Multimodal LLM-assisted text generation

To ensure stable pedestrian descriptions and mitigate linguistic noise that often arises from varying prompt formulations, we adopt a fixed-instruction strategy for prompting the LLM. Specifically, we utilize Qwen2.5-VL-7B as our generative backbone. Although a static prompt is used, the richness and diversity of the resulting pedestrian descriptions are not constrained by the instruction itself. Instead, they are dynamically driven by the actual visual content of each image through a closed-loop refinement mechanism. This strategy ensures that the generated text remains consistent in structure while being highly adaptive to the fine-grained attributes of different individuals. We design a generic instruction to ensure broad coverage of pedestrian attributes while minimizing spurious details:

"Please describe the overall appearance of the person in the image, including clothing (top, pants, shoes), hairstyle, gender, and carried items. If certain features are unclear, you may omit them, but do not include any information not actually present in the image."

Despite the rich semantic content of the generated text, cross-modal gaps can introduce detail inaccuracies such as incorrect colors or missing items. Hence, we employ a noise estimation algorithm based on CLIP to detect and refine these errors. First, we extract the visual feature V from the input image I , which is cropped from ground-truth instance bounding boxes, via CLIP's image encoder $f_{\text{img}}(\cdot)$:

$$V = f_{\text{img}}(I) \in \mathbb{R}^D. \quad (1)$$

Next, we tokenize the raw LLM-generated text T into $[w_1, w_2, \dots, w_N]$ using the BPE tokenizer from OpenCLIP and pass the full sentence through CLIP's text encoder $f_{\text{text}}(\cdot)$. The output representation at each token position is taken as t_i . Due to causal self-attention in the text encoder, each t_i is context-aware and encodes information from the surrounding tokens. We then compute the cosine similarity s_i between V and each t_i :

$$s_i = \frac{V \cdot t_i}{\|V\| \|t_i\|}. \quad (2)$$

Any token w_i whose similarity s_i falls below the threshold $\tau = 0.2$ is deemed erroneous, and its index is stored in the set $E = \{w_i \mid s_i < \tau\}$. While the original text remains intact, we mark these tokens for refinement. To enhance text accuracy, we feed the erroneous set E back into the LLM with an additional prompt:

"Please re-describe the person, ensuring that the clothing, colors, and accessories match the image. The following parts may be incorrect; please check and revise: {E}."

Formally, we obtain an updated description $T_{n+1} = \text{LLM}(I, P, T_n, E_n)$, where P is the prompt, and (T_n, E_n) represents the text and erroneous token set from the previous iteration. The LLM receives the flagged tokens together with the original image and re-evaluates them in their full linguistic context before deciding on revisions. This iterative refinement progressively aligns the text with the visual input and terminates when the erroneous set E becomes empty or when a maximum of two iterations ($n_{\max} = 2$) is reached.

3.2. Semantic structural consistency constraint

We utilize the refined textual features obtained from the previous text optimization module to guide the learning of visual representations. Specifically, we introduce a SSCC module, comprising two complementary loss terms: an instance-level matching loss and a global relation-level consistency loss.

Followed by IRRA (Jiang & Ye, 2023), we adopt the similarity distribution matching loss that encourages alignment between individual image-text pairs. Given visual feature f^v and textual feature f^t , the matching probability $p_{i,j}$ between the visual feature f_i^v and the textual feature f_j^t is computed as:

$$p_{i,j} = \frac{\exp(\text{sim}(f_i^v, f_j^t)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(f_i^v, f_k^t)/\tau)}, \quad (3)$$

where τ is a temperature parameter controlling distribution sharpness. To align the predicted distribution $p_{i,j}$ with the ground-truth probability $q_{i,j}$, we apply the Kullback-Leibler (KL) divergence as follows:

$$\mathcal{L}_{i2t} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N p_{i,j} \log\left(\frac{p_{i,j}}{q_{i,j} + \epsilon}\right), \quad (4)$$

where ϵ is a small constant added to maintain numerical stability, and the target probability $q_{i,j}$ is defined as one over the number of matched texts for the same instance and zero otherwise, which also applies to unlabeled pedestrians since their cropped boxes yield descriptions. By enforcing instance-level semantic alignment, this loss encourages fine-grained alignment between image and text representations, promoting discriminative visual features guided by textual semantics. This forward KL direction takes the predicted distribution as the first argument and the target as the second, which matches prior alignment frameworks and stresses correct matches.

However, focusing solely on instance-level alignment neglects the global relational structure among different pedestrian identities, potentially reducing the discriminative power of learned representations. To address this, we introduce a complementary global relation-level consistency loss, defined as:

$$\mathcal{L}_R = \frac{1}{N^2} \sum_{j=1}^N \sum_{k=1}^N (\text{sim}(f_j^v, f_k^v) - \text{sim}(f_j^t, f_k^t))^2. \quad (5)$$

This constraint enforces consistency between visual and textual embedding spaces by aligning their relational geometry across all pairwise interactions. Similar relation-level losses have been explored in vision-language modeling (Cao et al., 2024; Goel et al., 2022), but in our framework, we adapt it specifically for person search by introducing a normalized all-to-all consistency ($\frac{1}{N^2}$) instead of batch-level averaging. Different from usual instance wise alignment, the design forms a cross modal structural linkage that treats textual descriptions as steady

guides. Through capturing wider relational patterns between visual features and high level captions, it helps preserve semantic consistency and strengthen identity separability under complex variations.

3.3. Multimodal-aware self-distillation framework

Although the proposed instance-level matching and global relation-level consistency losses effectively align visual and textual representations, the learned model may still struggle to generalize to complex scenarios. To address this, we propose a MSDF, where both a frozen and a learnable CLIP-based encoder are employed. The frozen backbone, which benefits from CLIP's extensive pre-training on diverse web-scale datasets, is capable of extracting robust semantic features that generalize effectively to downstream pedestrian retrieval tasks. By applying this frozen model to cropped pedestrian instances, we provide stable multimodal priors that serve as a reliable reference for representation learning. Meanwhile, the learnable backbone operates on full-scene images to capture task-adaptive representations. Self-distillation is performed at both the instance and global relation levels, enabling the model to effectively internalize the stable semantic manifold and maintain consistent structural relationships among identities.

Specifically, we first obtain stable reference features from a frozen CLIP vision encoder for distillation. Given the frozen visual feature f_i^{v*} and textual feature f_j^t , we compute the multimodal matching probability:

$$p_{i,j}^* = \frac{\exp(\text{sim}(f_i^{v*}, f_j^t)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(f_i^{v*}, f_k^t)/\tau)}. \quad (6)$$

To enhance instance-level semantic consistency, we regularize the learnable encoder by minimizing the KL divergence between the current matching distribution $p_{i,j}$ and the frozen multimodal prior $p_{i,j}^*$:

$$\mathcal{L}_{i2t}^* = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N p_{i,j} \log\left(\frac{p_{i,j}}{p_{i,j}^* + \epsilon}\right). \quad (7)$$

Consequently, the final instance-level similarity distribution loss is defined as:

$$\mathcal{L}_{i2t}^{\text{total}} = \mathcal{L}_{i2t} + \gamma \mathcal{L}_{i2t}^*, \quad (8)$$

where γ balances the influence of the multimodal prior.

In parallel, we impose a global relation-level constraint through self-distillation, formulated as:

$$\mathcal{L}_R^* = \frac{1}{N^2} \sum_{j=1}^N \sum_{k=1}^N (\text{sim}(f_j^v, f_k^v) - \text{sim}(f_j^{v*}, f_k^{v*}))^2. \quad (9)$$

Different from conventional distillation that replicates features between trainable networks, our approach uses a fixed CLIP visual encoder pre-trained on large scale web multimodal data as the reference model. This yields persistent structural cues that assist the adapting visual backbone in absorbing broad cross modal knowledge. As a result, the model is less likely to overfit to confined environment dependent visual patterns and shows improved resilience to occlusion and cluttered backgrounds. Thus, the final global relation-level consistency loss is:

$$\mathcal{L}_R^{\text{total}} = \mathcal{L}_R + \delta \mathcal{L}_R^*, \quad (10)$$

where δ adjusts the influence of the global self-distillation constraint.

Together, $\mathcal{L}_{i2t}^{\text{total}}$ and $\mathcal{L}_R^{\text{total}}$ provide complementary supervision at the instance and relation levels, ensuring both fine-grained semantic alignment and stable global structure across modalities.

3.4. Training and inference

Our method builds upon a CLIP-ResNet-50 (Radford et al., 2021) backbone, denoted by ϕ_0 , to perform both person detection and person ReID within a unified framework. During training, we incorporate the

text generation module to provide additional supervision from large language models. This module is exclusively used in the training phase to refine textual descriptions and align them with the visual features. Once each mini-batch of images is processed, our network outputs bounding boxes and corresponding ReID features. We adopt a standard detection loss $\mathcal{L}_{det}^{total}$ that includes classification and regression components (e.g., cross-entropy for classification and Smooth- L_1 for bounding box regression). A fixed weight $\lambda = 10$ is applied to the regression term following the baseline setting.

$$\mathcal{L}_{det}^{total} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{reg}. \quad (11)$$

To provide effective supervision for the ReID task, we follow the OIM loss (Xiao et al., 2017), denoted by $\mathcal{L}_{ID}^{total}$. Specifically, we maintain a lookup table (LUT) M for all labeled identities and a circular queue U to store the features of unlabeled identities from recent mini-batches. At each training iteration, we compute the cosine similarities between current ReID features and all entries in M and U , yielding the probability of each sample belonging to a labeled or unlabeled identity. The OIM loss then encourages correct classification of each feature to its ground-truth identity while updating M and U accordingly.

Alongside detection and OIM losses, we integrate the proposed semantic correlation consistency losses, $\mathcal{L}_R^{total}$ and $\mathcal{L}_{i2i}^{total}$, to align visual features with CLIP text features and preserve inter-person relationships. Overall, the complete objective function is:

$$\mathcal{L} = \mathcal{L}_R^{total} + \mathcal{L}_{i2i}^{total} + \mathcal{L}_{ID}^{total} + \mathcal{L}_{det}^{total}, \quad (12)$$

where $\mathcal{L}_R^{total}$ and $\mathcal{L}_{i2i}^{total}$ are the regularized losses introduced in Sections 3.2 and 3.3, $\mathcal{L}_{ID}^{total}$ is the OIM loss, and $\mathcal{L}_{det}^{total}$ is the detection loss. The overall loss coefficients are all kept as 1. The internal coefficients of $\mathcal{L}_R^{total}$ and $\mathcal{L}_{i2i}^{total}$ are detailed above, while the OIM and detection losses follow the baseline settings.

During inference, we remove the text generation module and follow a standard two-stage process akin to many person search frameworks (Li & Miao, 2021). First, the trained model detects pedestrian bounding boxes in the input images, providing classification scores and bounding box coordinates. Second, for each detected pedestrian instance, we extract the learned ReID feature from our visual backbone. Person retrieval is then performed by comparing these features with the query person's representation typically via cosine similarity. Although textual guidance is absent at test time, the visual backbone has already internalized the fine-grained semantic manifold and inter-person relationships during the learning phase. Since the backbone is progressively aligned with the multimodal latent space of the frozen CLIP teacher, it learns to independently map pedestrian pixels into a semantically structured embedding space. This ensures that the feature distribution remains consistent across training and inference, allowing the model to maintain strong discriminative power and robustness using only visual cues without inducing distribution shifts.

4. Experiments and discussion

4.1. Datasets and implementation details

Datasets. We conduct evaluations on two widely used datasets to ensure fair comparisons. The CUHK-SYSU dataset (Xiao et al., 2017) comprises 18,184 images annotated with 8432 unique identities and 96,143 bounding boxes. It spans diverse environments, including urban streets and film scenes, with variations in viewpoints, lighting conditions, backgrounds, and pedestrian scales. Following the standard protocol, the training set includes 5532 identities and 11,206 images, while the test set consists of 2900 query persons and 6978 images. Validation results are reported across gallery sizes ranging from 50 to 4,000. The PRW dataset (Zheng et al., 2017) features imagery captured by six cameras, containing 932 distinct identities and 43,110 pedestrian annotations within 11,816 frames. The dataset is partitioned into a training set

Table 1

Analysis of Ablation Experiments on MLTG, SSCC, and MSDF on CUHK-SYSU and PRW datasets. * indicates that the ResNet50 backbone is based on CLIP.

Method	MLTG	SSCC	MSDF	PRW		CUHK-SYSU	
				mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
Backbone: ResNet50							
Baseline				46.7	83.4	93.8	94.6
+ SSCC		✓		49.7	83.9	94.1	94.8
+ SSCC + MSDF		✓	✓	53.2	85.4	94.6	95.1
+ MLTG + SSCC	✓	✓		51.2	84.9	94.4	95.0
MGSD	✓	✓	✓	55.2	87.6	94.9	95.3
Backbone: CLIP-ResNet50							
Baseline*				48.0	86.7	92.8	93.4
+ SSCC		✓		51.9	86.3	93.4	94.1
+ SSCC + MSDF		✓	✓	55.4	87.4	95.2	95.5
+ MLTG + SSCC	✓	✓		53.2	87.0	93.9	94.5
MGSD*	✓	✓	✓	56.1	88.4	95.5	96.0

Table 2

Analysis of Ablation Experiments on Individual Components of MLTG, SSCC, and MSDF on CUHK-SYSU and PRW datasets.

T	T_r	$\mathcal{L}_{i2i}$	$\mathcal{L}_R$	$\mathcal{L}_R^*$	$\mathcal{L}_{i2i}^*$	PRW		CUHK-SYSU	
						mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
						48.0	86.7	92.8	93.4
✓		✓				50.3	85.5	93.1	93.4
✓			✓			51.3	85.9	93.2	93.7
✓		✓	✓			51.9	86.3	93.4	94.1
✓		✓			✓	52.2	86.1	93.3	93.6
✓			✓	✓		53.3	86.9	94.3	95.5
✓		✓	✓	✓	✓	55.4	87.4	95.2	95.5
	✓	✓				50.9	85.7	93.3	93.7
	✓		✓			52.7	86.4	93.4	94.3
	✓	✓	✓			53.2	87.0	93.9	94.5
	✓	✓			✓	52.9	86.6	93.7	94.1
	✓		✓	✓		54.1	87.2	94.7	95.6
	✓	✓	✓	✓	✓	56.1	88.4	95.5	96.0

with 5704 images and 482 identities and a test set with 2057 query persons and 6112 images. All evaluations are based on this split, utilizing the clear metrics (Xiao et al., 2017; Zheng et al., 2017), including mAP and Top-1 accuracy.

Implementation Details. Following previous multimodal works such as Bai et al. (2023), Wu et al. (2023), Yang et al. (2023b), the LLM-based caption generation is conducted once in an offline manner prior to training, introducing no additional overhead during training and inference. The generated textual and CLIP features are extracted offline and directly utilized in the training phase. Subsequently, we build upon prior works (Cao et al., 2022; Chen et al., 2020c; Li & Miao, 2021; Yu et al., 2022) and adopt ResNet50 (He et al., 2016) as the backbone. Training is performed using the SGD optimizer with a momentum of 0.9 for 15 epochs. The initial learning rate gradually increases to 0.001 during the first epoch, then decreases by a factor of 10 at epochs 10 and 14. No further augmentation is applied beyond the baseline configuration and all primary experiments adopt a fixed random seed of 37 to ensure reproducible results. Consistent with Yu et al. (2022), the model is trained using a single NVIDIA RTX 4090 GPU with a batch size of 8 images and tested on input images resized to 1500×900 pixels.

4.2. Ablation studies

Effectiveness of Different Modules. Tables 1 and 2 summarize the ablation results at both module and fine-grained levels. As shown in the module-level analysis (Table 1), starting from the baseline (48.0% mAP on PRW under the CLIP-based backbone), the contribution of SSCC reaches 51.9% mAP. By integrating MSDF on top of the structural foun-

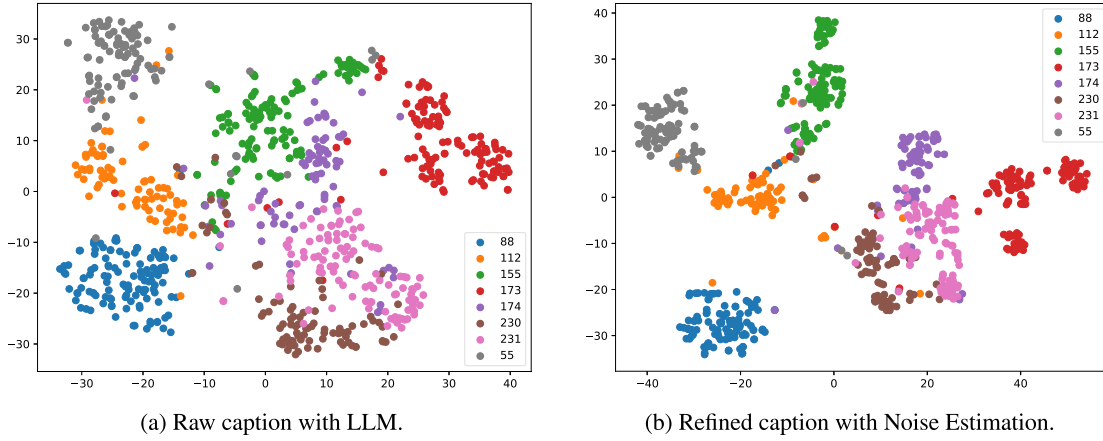


Fig. 3. Visualization of caption quality improvement through noise estimation.

Table 3

Ablation studies of loss replacements and detector crops on PRW and CUHK-SYSU datasets.

Method	$\mathcal{L}_R$	$\mathcal{L}_R^*$	GT crops	PRW		CUHK-SYSU	
				mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
$\mathcal{L}_{sim}$		✓	✓	53.3	86.9	93.9	94.2
$\mathcal{L}_{mse}^*$	✓		✓	52.9	86.4	93.2	94.0
T_{det}	✓	✓		54.1	87.0	94.2	95.1
MGSD*	✓	✓	✓	56.1	88.4	95.5	96.0

dation of SSCC, the performance significantly increases to 55.4% mAP. Finally, incorporating MLTG to refine the textual supervision leads to our state-of-the-art result of 56.1% mAP. Notably, the backbone ablation in Table 1 demonstrates that our modules are equally effective on a standard ResNet50, where MGSD elevates the baseline from 46.7% to 55.2% mAP. This steady performance climb confirms the necessity and synergistic contribution of each macro-component. To further disentangle these gains, the fine-grained breakdown in Table 2 provides a detailed analysis of our framework. Specifically, the comparison between raw (T) and refined (T_r) descriptions in MLTG confirms that CLIP-based noise estimation effectively mitigates LLM hallucinations, ensuring more precise semantic alignment. Furthermore, the SSCC module structures the feature space through synergistic alignment at both the instance-matching ($\mathcal{L}_{i2i}$) and global structural consistency ($\mathcal{L}_R$) levels. Ultimately, the self-distillation terms ($\mathcal{L}_{i2i}^*$, $\mathcal{L}_R^*$) in MSDF, when integrated with the existing structural base, significantly boost the robustness of the model against real-world variations such as changes in illumination and occlusion. We further examine the loss formulations in SSCC and MSDF. As shown in Table 3, replacing the caption-aware relational constraint with a standard similarity-preserving distillation loss $\mathcal{L}_{sim}$ leads to a clear performance drop, and substituting the relation-level self-distillation with ordinary feature-level MSE loss $\mathcal{L}_{mse}$ similarly degrades results. Both reductions confirm that the relational geometry, rather than simple feature mimicry, drives the distillation gain. Overall, these modules work complementarily to construct a unified and discriminative representation for person search.

Impact of Text Refinement. In this ablation study, we investigate the impact of incorporating our CLIP-based iterative noise detection and refinement procedure on the quality of language descriptions. Fig. 3a and b present t-SNE visualizations of the textual embeddings generated without and with our refinement pipeline, respectively. In Fig. 3a, the raw LLM-generated descriptions form more dispersed distributions, with substantial overlap among clusters corresponding to different person identities. This dispersion indicates inconsistencies or inaccuracies in the originally generated text, such as misidentified colors or missing at-

Table 4

Comparison of noise-estimation granularity on PRW and CUHK-SYSU datasets. Caption-level regenerates the whole description without localized error signals, while token-level localizes erroneous tokens before refinement.

Granularity	PRW		CUHK-SYSU	
	mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
Caption-level	55.7	87.9	95.3	95.7
Token-level	56.1	88.4	95.5	96.0

tributes, which lead to suboptimal cross-modal alignment. By contrast, the refined embeddings in Fig. 3b exhibit more coherent clustering and clearer inter-cluster boundaries, demonstrating improved consistency across pedestrian descriptions. This result highlights the effectiveness of iteratively detecting erroneous tokens via CLIP similarity and prompting the LLM for targeted revisions. Critically, the iterative mechanism prevents repeated propagation of incorrect information, leading to more faithful representations of clothing, color, and other fine-grained attributes. Consequently, the refined text descriptions align more closely with the associated image features in the joint vision-language embedding space, thereby offering more robust semantic representations for downstream tasks such as person ReID or cross-modal retrieval. These qualitative improvements are supported by quantitative results in Table 2, where incorporating refined textual descriptions increases mAP from 55.4% to 56.1% and Top-1 accuracy from 87.4% to 88.4% on PRW, confirming that the refinement module effectively enhances cross-modal feature alignment. Table 3 also reports the T_{det} variant, where captions are generated from detector crops produced by an external frozen Faster R-CNN instead of GT crops. Compared with MGSD*, this yields a moderate gap of 2.0% mAP on PRW and 1.3% mAP on CUHK-SYSU, yet T_{det} still surpasses all prior methods on both datasets. This indicates that the high-level attributes captured in the textual descriptions remain robust under typical localization errors from a realistic detector. To further validate the granularity of the proposed noise-estimation strategy, we compare token-level refinement with a caption-level regeneration variant, where the whole description is regenerated without localized error signals. As shown in Table 4, token-level refinement consistently outperforms caption-level regeneration on both datasets. This indicates that localized erroneous-token cues provide more precise guidance for caption correction than regenerating the entire sentence.

Impact of Noise Threshold. To assess the impact of the noise threshold τ on pedestrian descriptions, we conducted experiments on the CUHK-SYSU and PRW datasets (Table 5). As τ increases from 0.0 to 0.2, performance improves, reaching the highest mAP and Rank-1 accuracy at $\tau = 0.2$. This indicates that smaller thresholds failed to effectively

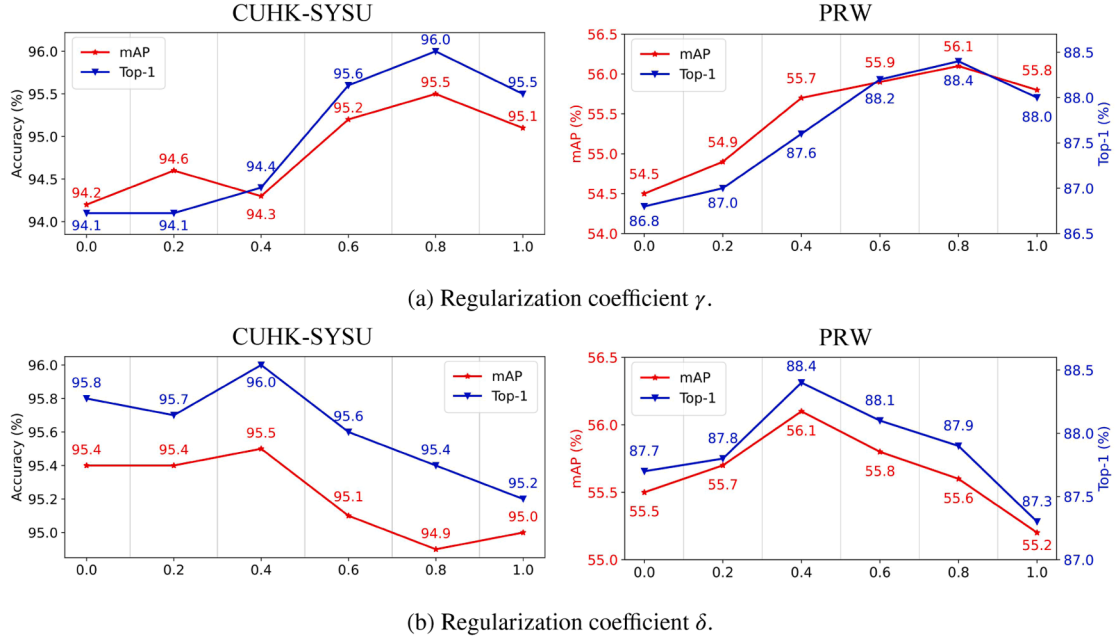


Fig. 4. The contribution of the regularization coefficients γ and δ in the detection context.

Table 5

Performance comparison of noise threshold τ on CUHK-SYSU and PRW datasets.

τ	CUHK-SYSU		PRW	
	mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
0.0	95.2	95.5	55.4	87.4
0.1	95.4	95.8	55.8	88.0
0.2	95.5	96.0	56.1	88.4
0.3	95.3	95.6	55.6	87.9
0.4	95.0	95.3	55.2	87.1

remove erroneous tokens, while larger ones (e.g., 0.3 and 0.4) resulted in over-refinement, negatively affecting text quality. The experiments show that $\tau = 0.2$ strikes the best balance between error detection and preserving accurate descriptions, proving to be optimal across different settings.

Impact of Regularization Coefficients. We analyze the influence of regularization coefficients γ and δ in our multimodal self-distillation framework through an ablation study on both CUHK-SYSU and PRW datasets (Fig. 4), following the standard person search evaluation protocol where mean Average Precision (mAP) and Top-1 accuracy are reported. As illustrated in Fig. 4a, increasing γ initially boosts mAP and Top-1 accuracy by effectively aligning visual-textual features via CLIP’s vision encoder, which introduces valuable semantic guidance. However, performance deteriorates when $\gamma > 0.8$, likely because excessive regularization constrains the model’s flexibility and reduces adaptability to visual variations. Similarly, Fig. 4b shows that a moderate δ (around 0.4) enhances inter-person relationship modeling and improves discrimination between identities, while higher values ($\delta > 0.6$) hinder task-specific feature learning. These trends are consistent across both datasets, indicating that the regularization strategy is dataset-agnostic and can be generalized to other datasets with appropriate tuning of γ and δ .

Convergence Analysis. To analyze the convergence behavior of our method, we evaluate performance across different numbers of training epochs while keeping all other settings unchanged. As shown in Table 6, the performance improves steadily from 5 to 15 epochs, indicating that additional training iterations enhance feature discrimination and

Table 6

Performance under different training epochs on CUHK-SYSU and PRW datasets.

epoch	CUHK-SYSU		PRW	
	mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
5	93.7	94.0	48.7	85.5
10	95.2	95.7	55.4	87.3
15	95.5	96.0	56.1	88.4
20	95.1	95.4	53.0	86.9
25	94.9	95.1	51.6	86.1

Table 7

Evaluation of model generalization capability. Results for SeqNet* and MGSD* are reported as mean $\pm$ std over 5 independent runs. The models are trained on CUHK-SYSU and tested on PRW (left) or trained on PRW and tested on CUHK-SYSU (right). * indicates that the ResNet50 backbone is based on CLIP.

Method	Target: PRW		Target: CUHK-SYSU	
	mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
SeqNet (Li & Miao, 2021)	21.2	65.4	42.3	44.7
SeqNet* (Li & Miao, 2021)	30.86 $\pm$ 0.34	77.95 $\pm$ 0.27	44.91 $\pm$ 0.38	50.73 $\pm$ 0.29
MGSD*	36.61 $\pm$ 0.13	81.07 $\pm$ 0.16	54.74 $\pm$ 0.11	58.08 $\pm$ 0.15

optimization stability. After 15 epochs, the accuracy begins to decline slightly, suggesting overfitting to the training data. This trend demonstrates that our model can effectively exploit the available data within a moderate training range, while excessive training may reduce its generalization ability.

Cross-Dataset Generalization Performance. Table 7 evaluates the generalization capability of different methods by testing models trained on CUHK-SYSU directly on PRW and vice versa, without any fine-tuning on the target dataset. To ensure statistical reliability, all results for SeqNet* and MGSD* are reported as the mean $\pm$ standard deviation over five independent runs. Our method consistently achieves superior performance across both scenarios, demonstrating its strong ability to generalize to unseen domains. When trained on CUHK-SYSU and evaluated on PRW, our method achieves 36.61% $\pm$ 0.13% mAP and 81.07% $\pm$ 0.16% Top-1 accuracy, significantly outperforming SeqNet (Li & Miao, 2021). Similarly, when trained on PRW and tested on CUHK-SYSU, our

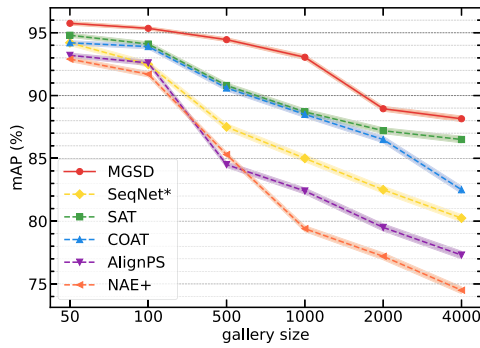


Fig. 5. Comparative evaluation of person search performance on the CUHK-SYSU dataset under varying gallery sizes. The curves and shaded areas denote the mean mAP and standard deviation across five independent runs, respectively.

model attains $54.74\% \pm 0.11\%$ mAP and $58.08\% \pm 0.15\%$ Top-1 accuracy, highlighting its robustness in adapting to diverse datasets with varying scene distributions. Notably, SeqNet* employs a CLIP-based ResNet50 backbone and shows improved performance compared to the standard SeqNet, confirming the benefits of leveraging vision-language pretraining for more transferable representations. However, our method surpasses SeqNet* in both cross-domain settings, indicating that the performance gains are not merely attributed to the backbone but also to the enhanced feature learning and domain alignment strategies. These results underscore the effectiveness of our approach in real-world person search applications, where models must generalize across different environments without dataset-specific tuning.

4.3. Comparison with state-of-the-art

As presented in Table 8, we compare our MGSD with state-of-the-art algorithms, including both two-step methods and end-to-end methods, on CUHK-SYSU and PRW datasets.

Results on CUHK-SYSU. As summarized in Table 8, our proposed method achieves a new state-of-the-art performance on the CUHK-SYSU benchmark with a gallery size of 100. Specifically, it attains an mAP of 95.5% and a Top-1 accuracy of 96.0%, outperforming all prior methods under the same ResNet-50 backbone. Compared to recent strong methods such as UGAS (Li et al., 2024b) (94.3% mAP, 94.8% Top-1), GMT (Cai et al., 2025) (94.8%, 95.4%), and PSDiff (Jia et al., 2025a) (95.1%, 95.3%), our approach consistently outperforms them by a clear margin of 0.4–1.2% in mAP and 0.6–1.2% in Top-1 accuracy. These improvements highlight the effectiveness of our proposed framework in enhancing retrieval quality under both precision and ranking metrics. In addition, we re-evaluate representative open-source baselines (e.g., COAT and SAT) with a CLIP-ResNet50 backbone following the same protocol in Section 4.1. Directly replacing the standard ImageNet-pretrained backbone with a CLIP-based one leads to a performance drop for these vision-only methods, indicating that they cannot fully exploit the semantically aligned feature space of CLIP without dedicated adaptation mechanisms. To comprehensively evaluate the robustness and scalability of our approach, we further conduct experiments under varying gallery sizes, ranging from 50 to 4,000. This setting reflects the diversity of real-world deployment scenarios, where the number of candidate images can vary drastically. As shown in Fig. 5, where the curves represent the mean performance over five independent runs and the shaded areas denote the standard deviation, our method maintains consistently high performance across different gallery sizes, demonstrating strong generalization ability. The effectiveness stems from a more robust feature representation, an optimized detection-retrieval synergy, and improved adaptation to large-scale scenarios, enabling more stable person search performance even in challenging conditions.

Table 8

Comparison with state-of-the-art methods in terms of mAP and Top-1 accuracy on the CUHK-SYSU and PRW test sets. All methods are implemented with the ResNet50 backbone. * indicates that the ResNet50 backbone is based on CLIP.

Method	CUHK-SYSU		PRW	
	mAP(%)	Top-1(%)	mAP(%)	Top-1(%)
Two-Step				
DPM (Zheng et al., 2017)	–	–	20.5	48.3
MGTS (Chen et al., 2018)	83.0	83.7	32.6	72.1
CLSA (Lan et al., 2018)	87.2	88.5	38.7	65.0
RDLR (Han et al., 2019)	93.0	94.2	42.9	70.2
IGPN (Dong et al., 2020b)	90.3	91.4	47.2	87.0
TCTS (Wang et al., 2020)	93.9	95.1	46.8	87.5
One-step				
OIM (Xiao et al., 2017)	75.5	78.7	21.3	49.4
QEEPS (Munjal et al., 2019)	88.9	89.1	37.1	76.7
HOIM (Chen et al., 2020a)	89.7	90.8	39.8	80.4
BiNet (Dong et al., 2020a)	90.0	90.7	45.3	81.7
NAE (Chen et al., 2020c)	91.5	92.4	43.3	80.9
DMRNet (Han et al., 2021b)	93.2	94.2	46.9	83.3
DKD (Zhang et al., 2021)	93.1	94.2	50.5	87.1
PGS (Kim et al., 2021)	92.3	94.7	44.2	85.2
AlignPS (Yan et al., 2021)	93.1	93.4	45.9	81.9
OIMNet++ (Lee et al., 2022)	93.1	93.9	46.8	83.9
PSTR (Cao et al., 2022)	93.5	95.0	49.5	87.8
COAT (Yu et al., 2022)	94.2	94.7	53.3	87.4
DMRNet++ (Han et al., 2022)	94.5	95.7	52.1	87.0
SeqNet (Li & Miao, 2021)	94.1	94.7	50.8	86.0
SAT (Fiaz et al., 2023)	94.4	94.8	54.5	87.5
ROI-AlignPS (Yan et al., 2023)	95.4	96.0	51.6	84.4
IAM (Tian et al., 2024)	95.4	96.0	54.5	87.6
PAD (Kim et al., 2024)	94.6	95.4	54.8	88.2
UIL (Yan & Li, 2024)	93.9	94.7	51.5	86.1
UGAS (Li et al., 2024b)	94.3	94.8	51.9	85.5
GMT (Cai et al., 2025)	94.8	95.4	51.3	87.0
PSDiff (Jia et al., 2025a)	95.1	95.3	53.5	87.1
SeqNet (Li & Miao, 2021)	93.8	94.6	46.7	83.4
SeqNet* (Radford et al., 2021)	92.8	93.4	48.0	86.7
COAT* (Yu et al., 2022)	93.5	94.2	51.3	86.0
SAT* (Fiaz et al., 2023)	93.7	94.1	52.1	86.6
MGSD*	95.5	96.0	56.1	88.4

Table 9

Person search results for MGSD and four compared methods on PRW. The performance is decomposed into search results under detected boxes (Top) and pure ReID performance under ground-truth boxes (Bottom).

Method	ReID		Detection	
	mAP(%)	Top-1(%)	Recall(%)	AP(%)
Evaluated under detected boxes				
NAE (Chen et al., 2020c)	43.3	80.9	93.3	88.8
SeqNet (Li & Miao, 2021)	46.7	83.4	95.4	93.3
COAT (Yu et al., 2022)	53.3	87.4	94.3	92.6
SAT (Fiaz et al., 2023)	54.5	87.5	94.9	92.8
MGSD	56.1	88.4	96.3	93.5
Evaluated under ground-truth boxes				
NAE (Chen et al., 2020c)	46.9	80.9	100	100
SeqNet (Li & Miao, 2021)	47.9	83.4	100	100
COAT (Yu et al., 2022)	54.7	87.4	100	100
SAT (Fiaz et al., 2023)	55.7	88.2	100	100
MGSD	57.6	88.9	100	100

Results on PRW. Table 8 presents the comparison of our method with state-of-the-art approaches on the PRW dataset, a more challenging benchmark characterized by larger scene variation and frequent occlusions. Our method achieves an mAP of 56.1% and a Top-1 accuracy of 88.4%, surpassing all existing methods. Notably, several recent approaches such as ROI-AlignPS (Yan et al., 2023) (51.6%, 84.4%), IAM

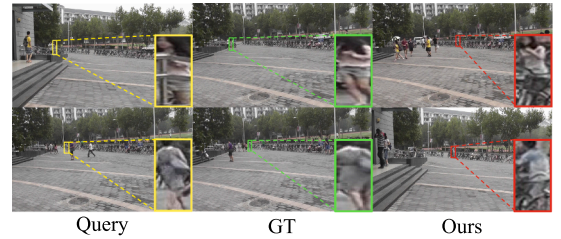


Fig. 6. Qualitative results on the CUHK-SYSU test set. The query pedestrian is indicated by the yellow bounding box, while red and green bounding boxes correspond to incorrect and correct rank-1 retrieval results, respectively.

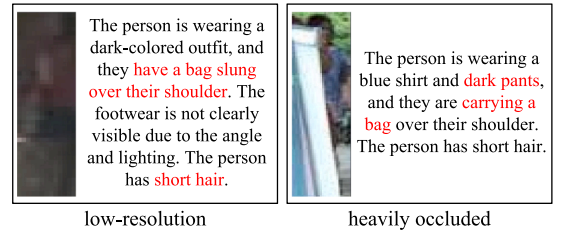
(Tian et al., 2024) (54.5%, 87.6%), and GMT (Cai et al., 2025) (51.3%, 87.0%) achieve comparable performance on CUHK-SYSU but are outperformed by our method on PRW. In contrast, our method maintains strong retrieval performance across both benchmarks, which demonstrates the effectiveness of our approach in handling real-world challenges in person search, where person detection and ReID must be jointly optimized. To provide a more detailed analysis under different evaluation settings, Table 9 provides an extended performance decomposition. In the end-to-end person search pipeline, our framework simultaneously achieves a high detection Recall of 96.3% and an AP of 93.5%, demonstrating its superior capability in accurately localizing pedestrians. Evaluating with ground-truth (GT) person boxes further eliminates the influence of detection errors, allowing for a more focused evaluation of the ReID component. Under this protocol, our method achieves an mAP of 57.6% and a Top-1 accuracy of 88.9%, indicating the strength of the learned feature representations and the effectiveness of our matching strategy.

Efficiency Comparison. Table 10 presents the efficiency comparison of different person search methods in terms of model size, computational complexity, inference cost, and training cost. Our MGSD achieves the highest mAP (56.1%) and Top-1 accuracy (88.4%) while maintaining a favorable balance between accuracy and efficiency. Specifically, despite involving slightly higher parameter counts (49.27 M) and MACs (286.37 G) compared to COAT*, our approach achieves superior accuracy and nearly twice the inference speed (21.47 FPS vs. 10.73 FPS), demonstrating its superior throughput efficiency. In terms of training cost, MGSD requires a moderate average training time per epoch (15.1 minutes), which is comparable to the SeqNet baseline and notably lower than COAT. As for offline caption generation cost, this one-time pre-processing step (e.g., 8 hours for PRW and 26 hours for CUHK-SYSU) is incurred exclusively prior to training and adds zero burden to the inference phase. The total wall-clock duration from data preparation to final optimization reaches roughly 11.8 hours on PRW and 29.8 hours on CUHK-SYSU in our GPU setting, yet this upfront cost does not alter deployment efficiency. We also observe that transitioning from standard ResNet50 to a CLIP-based backbone results in minimal changes to the model’s static complexity. These results highlight the effectiveness of our design in optimizing the trade-off between discriminative power and computational efficiency, making it well-suited for real-world applications where real-time inference is critical.

Qualitative Results. Fig. 6 presents qualitative comparisons of our method (MGSD) against AlignPS (Yan et al., 2021), PSTR (Cao et al., 2022), and SAT (Fiaz et al., 2023) on the CUHK-SYSU test set. The yellow boxes denote query persons and red boxes present the wrong results, while the green boxes indicate the correctly retrieved matches. Un-



(a) Failure cases on the PRW test set. The query pedestrian is indicated by the yellow bounding box, while green and red bounding boxes correspond to ground truth and incorrect rank-1 retrieval results, respectively.



(b) Failure cases of LLM-generated descriptions. Red-marked regions indicate parts of the description that are erroneous or uncertain, due to the limited visual evidence.

Fig. 7. Failure cases of the proposed method.

der partial occlusion (row 1), high inter-class visual similarity (row 2), and crowded scenes combining both factors (row 3), only our semantic-aware, multimodal-guided model consistently retrieves the true target. This demonstrates the model’s ability to leverage rich semantic cues beyond pure visual appearance, effectively disambiguating confusing cases. In contrast, existing methods often struggle in cluttered environments, frequently misidentifying background distractors or visually similar identities. Notably, in the first and third rows, our method successfully retrieves the target person with high precision and confidence, while others fail to do so or produce multiple false positives. For completeness, we also show several failure cases on the PRW dataset in Fig. 7a, where our method is confused by extreme small targets. As can be seen, our method is unable to retrieve the corresponding query person when they are diminutive. These examples indicate situations where limited visual information reduces the effectiveness of semantic guidance, leading to misretrieval.

Table 10

Comparison of person search efficiency, including both training and inference efficiency. The proposed MGSD needs a one time offline caption stage lasting 8 hours on PRW and 26 hours on CUHK-SYSU.

Method	Params (M)	MACs (G)	Speed (FPS)	Training Time (mins / epoch)	mAP (%)	Top-1 (%)
NAE (Chen et al., 2020c)	33.43	287.35	21.23	6.2	43.3	80.9
AlignPS (Yan et al., 2021)	42.18	189.98	16.39	12.7	45.9	81.9
SeqNet (Li & Miao, 2021)	48.41	275.11	20.86	14.8	46.7	83.4
COAT (Yu et al., 2022)	36.95	236.25	11.24	23.1	53.3	87.4
COAT* (Yu et al., 2022)	36.97	241.52	10.73	24.3	51.3	86.0
SAT* (Fiaz et al., 2023)	34.34	291.62	14.51	18.5	52.1	86.6
MGSD	49.27	286.37	21.47	15.1	56.1	88.4

4.4. Limitation and future directions

Although MGSD achieves strong performance in complex environments, it still faces limitations in extremely challenging cases. As shown in Fig. 7b, when the pedestrian is heavily occluded, severely blurred, or captured at very low resolution, the generated textual descriptions may be incomplete or partially inaccurate due to the inherent hallucination tendency of LLMs, resulting in suboptimal semantic alignment. Additionally, MGSD may struggle with extremely small or distant pedestrians, where visual cues are limited and fine-grained feature extraction is difficult. Future work aims to investigate adaptive semantic reconstruction strategies that enhance robustness under modality imbalance or degraded inputs. Moreover, we plan to extend the MGSD framework to a wider range of multimodal tasks, such as text-image ReID or fine-grained scene understanding, while exploring the integration of more generalized foundation-model-based architectures to further strengthen cross-domain adaptability.

5. Conclusion

In this work, we propose a novel multimodal approach to address the limitations of existing person search methods, which often suffer from discrete identity representations and limited generalization. By leveraging textual descriptions as semantic priors, our method enhances pedestrian representations beyond conventional one-hot encoding, capturing richer inter-person relationships. To further reinforce feature discriminability, we introduce a semantic correlation consistency loss, ensuring structured constraints in the learned feature space while preserving meaningful similarities among visually similar individuals. Additionally, we incorporate CLIP-based self-distillation, aligning the learnable visual encoder with multimodal priors to improve robustness against illumination changes, occlusions, and background variations. Extensive experiments on benchmark datasets demonstrate that our approach significantly outperforms state-of-the-art methods, validating its effectiveness and potential for real-world applications. While promising, our approach may still face challenges in handling heavily occluded, low-resolution, or extremely small pedestrians, where textual cues become less reliable. Future research would explore adaptive text filtering and multi-scale visual enhancement to further mitigate these limitations.

CRedit authorship contribution statement

Xi Yang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition; **Hexun Zhou:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis; **Haiyang Zhu:** Writing – original draft, Visualization, Software, Methodology; **Nannan Wang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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